PROGRAM 7 – BREADTH FIRST SEARCH (BFS)

AIM

To write a Python program to implement the Breadth First Search algorithm to traverse a graph.

ALGORITHM

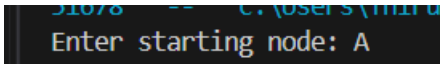
1. Start the program.
2. Create a graph using an adjacency list.
3. Read the starting node from the user.
4. Initialize an empty queue.
5. Insert the starting node into the queue.
6. Mark the starting node as visited.
7. Remove the first node from the queue.
8. Display the node.
9. Add all unvisited adjacent nodes to the queue.
10. Repeat until the queue becomes empty.
11. Display the BFS traversal order.

CODE

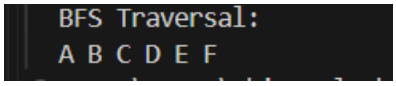
```
from collections import deque
graph={"A":["B","C"],"B":["D","E"],"C":["F"],"D":[],"E":["F"],"F":[]}
def bfs(start):
    visited=set()
    queue=deque([start])
    visited.add(start)
    print("BFS Traversal:")
    while queue:
        node=queue.popleft()
        print(node,end=" ")
        for neighbor in graph[node]:
            if neighbor not in visited:
                visited.add(neighbor)
                queue.append(neighbor)
start=input("Enter starting node: ").upper()
if start in graph:
```

```
    bfs(start)
else:
    print("Invalid starting node")
```

INPUT



OUTPUT



RESULT

Thus, the Breadth First Search algorithm was successfully implemented in Python to traverse the graph.